Attention-Sink Signatures Dissociate During Vision–Language Pretraining

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Abstract

Attention sinks are a widely studied target of interventions in language models and are often linked, though not causally, to hallucinations in vision-language models. A sink is defined by attention concentration ($\mathrm{Sink}_1^{\epsilon}$), but in text LMs it reliably arrives with two companions: a drained value norm (v-ratio) and massive activations (h-ratio) at the same position. We track all three separately during multimodal pretraining, without assuming they are one phenomenon. The model is a 222M vision-language model with a randomly initialized decoder, position 0 being an image token, no BOS, trained under four levers: softmax attention, output-gated softmax, unnormalized sigmoid attention, and initialization from a pretrained text LM. The three signatures dissociate: each training lever produces a different combination of them, consistently across 2-3 seeds per arm. On a 1B fresh-token run (2.39 effective visual epochs, single seed), the massive-activation proxy more than doubles while attention concentration stays at exactly zero at every threshold we test. Value-norm drain emerges as a third axis, beyond the massive-activation vs. sink dissociations known from text-only models.

1 Introduction

Decoder-only transformers pick up a habit early in training. A large share of attention goes to the first token or tokens of a sequence, whatever they contain. Streaming inference depends on that habit [1], the activation outliers that ride along with it make quantization harder [2], and a survey now organizes a subfield around interpreting and removing it [3].

The sink does not arrive alone. Three measurements move together in text language models. Attention concentrates on the sink token. The value vector there drops to a near-zero norm, called value-state drain [4]. The residual stream there grows an abnormally large norm, called a massive activation [2, 5]. These effects settle on the same few tokens [6] and appear near step 1000 [7], and the field reads the co-occurrence as permission to treat all three as facets of one attention-sink phenomenon.

Whether they really are one phenomenon is contested. One line argues for causal unity: massive activations mathematically require representational compression, and ablating them removes both compression valleys and sink formation [7], while a related view holds that outlier-driven rescaling by attention and residual sinks is essential to stable training [8]. Two recent text-only studies pull the other way, each separating one pair of the signatures, by normalization scheme [9] or by value scale [10]. Neither separates all three.

Which signature a study measures decides what it can conclude. If the three move independently, a lever that clears concentration while leaving the value path alone reads as a fix under one metric and a failure under the next, and two papers can disagree while both are right. Sink-like attention has also been tied to hallucination and weak visual grounding in deployed vision–language models [13, 14, 15, 16], so a mitigation credited with removing the sink may leave the signature that mattered untouched. We measure all three at once, throughout training, and test no downstream behavior.

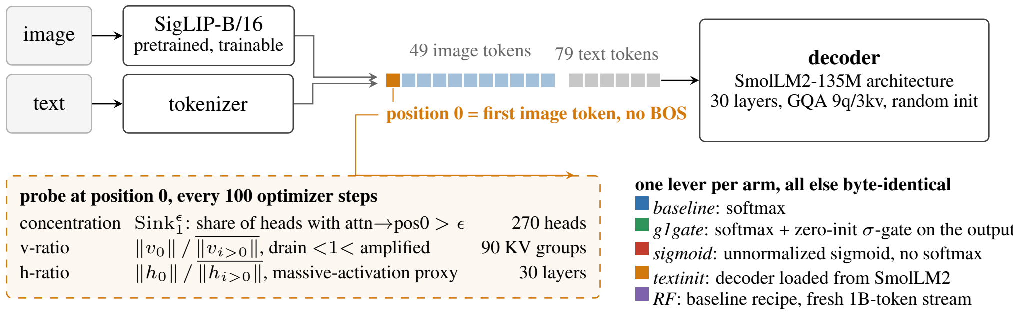

Figure 1: **Setup.** A pretrained SigLIP-B/16 encoder feeds 49 image tokens, then 79 text tokens, into a randomly initialized decoder of the SmolLM2-135M architecture. There is no BOS token, so position 0 is the first image token. A probe reads three signatures there every 100 steps, each at its own granularity. Four levers change one thing each, and RF repeats the baseline recipe on a fresh 1B-token stream.

Vision–language models suit the question, because the multimodal setting changes what position 0 *is*. Our sequence starts with 49 image tokens and has no BOS token, so the candidate sink token is a visual token, without the special-token machinery that text-LM sink accounts lean on. The multimodal sink studies we know analyze mostly frozen, already-trained backbones at inference time [11, 12], and the one training-time view follows a single magnitude rather than three signatures (Section 2). Telling whether the signatures arrive together, in sequence, or independently takes a randomly initialized decoder with all three logged separately from step 0.

We work at small scale by choice. The model is a 222M-parameter nanoVLM [17] with a pretrained SigLIP-B/16 encoder [18] and a randomly initialized decoder of the SmolLM2-135M architecture [19], trained under four levers that each target one sink-relevant mechanism (Section 3): softmax attention (*baseline*), a zero-initialized Qiu-style G1 output gate [20] (*g1gate*), unnormalized sigmoid attention, which removes the normalization sinks are argued to come from [6] (*sigmoid*), and initialization from the pretrained SmolLM2 text model (*textinit*). A validated probe logs all three signatures every 100 steps. Because the four-arm comparison reuses a small image pool, we re-test the central negative result on a fresh stream of one billion tokens (*RF*, 2.39 effective visual epochs).

This paper reports three observations from that setup. Decoupling itself is not new [9, 10] and we do not claim it. What is new is the conjunction: all three signatures tracked jointly, from step 0, under randomly initialized decoders in a multimodal model, with value-norm drain as a third axis. First, across four levers (n = 2–3 seeds per arm) the arms reach four different corners of the three-signature space, no two sharing a triple, and the value-norm ratio alone is strongly drained, mildly drained or amplified depending on the lever (Fig. 2, Table 1). The arms are not a factorial design, so these are intervention-associated profiles rather than isolated causal effects. Second, on a fresh stream at 2.39 effective visual epochs, with training healthy throughout (Section 3), the massive-activation proxy rises from an h-ratio of 1.43 to 3.22, about 2.3×, while concentration stays at exactly zero and no head ever crosses the sink threshold (Sections 4.2 and 5, one seed). Third, the per-head correlation between concentration and value-norm flips sign across arms at seed 0, +0.76 in baseline to −0.79 in textinit with a pooled −0.20, over 90 KV groups per arm, reported descriptively (Section 4.3).

2 Related work

Sinks and their companions in text language models Gu et al. [6] give the canonical account. The sink token acts “more like key biases, storing extra attention scores,” carries small key and value norms, and connects to massive residual-stream activations [2, 5]. They also show that unnormalized sigmoid attention prevents sink formation in text models up to 1B parameters, which our *sigmoid* arm builds on. Guo et al. [4] call the concentration and value-drain coupling “active-dormant” heads. Queipo-de-Llano, Arroyo et al. [7] make the strongest unity claim, and the only causal one. Ablating a model’s layer-0 massive activation removes both compression valleys and sink formation (Section 1). We reserve “causally unified” for that result. Peng et al. [24] trace a first-position sink circuit emerging early in from-scratch text pretraining, without separating the signatures.

Prior decoupling results, text-only, two axes Two papers already separate pairs of these signatures, and we scope our claim around them. Sun, Canziani, LeCun and Zhu [9] show that a change of normalization scheme crushes the massive-activation spike while the sink ratio survives. Chen and Yao [10] go the other way, from scratch. A value-scale intervention in 0.1–0.3B text models keeps the sinks and suppresses massive activations. Neither treats value-norm drain as a third axis, and neither is multimodal. Fesser et al. [23] give the closest support for tracking the value norm separately. One sink pattern can hide an "adaptive nop" with negligible value norms or a "broadcast" that redistributes global information, though that work diagnoses trained vision transformers rather than tracking emergence. Qiu et al. [8] argue the opposite case, that outlier-driven rescaling by attention and residual sinks is essential to stable training. The field has settled neither question.

Multimodal sinks, mostly frozen backbones at inference time Luo et al. [11] separate ViT-propagated from LLM-emerged sinks. Their Appendix A.4 tracks sink-dimension magnitudes across alignment checkpoints, so a training-time view has precedent, but on a frozen vision transformer with a pretrained language model, following one magnitude rather than three signatures. Choi et al. [12] likewise distinguish vision-sinks from language-sinks in a frozen model and gate them by layer. Both establish distinct vision-side and language-side origins, which our *textinit* inheritance result fits. Neither gives dense, joint tracking of the three quantities in a decoder that starts from random weights. Vision transformers also grow high-norm "register" tokens of their own [25], hence the pretrained-encoder limitation in Section 5. A separate line ties these signatures to hallucination and grounding failure in deployed models and intervenes at inference time [13, 14, 15, 16], the practical reason it matters which signature a mitigation moves.

The gating lever Qiu et al. [20] introduce the head-specific elementwise sigmoid gate on attention output that our *g1gate* arm adapts. In text models it "largely reduces the attention score allocated to the first token and decreases massive activations" while improving quality. Our zero-initialized variant confounds gating with initial output scaling (Sections 3 and 5). With that caveat, concentration is already absent in our baseline, so the gated arm differs on the value-norm axis (the v-ratio of Section 3). The drain becomes milder, 0.69–0.72 to 0.81–0.85, rather than disappearing, a difference invisible unless the three signatures are logged separately.

Positioning The closest prior work separates at most two of the three axes, in text-only models, and the multimodal studies work on frozen ones. Our claim is the conjunction. We track concentration, value-norm drain and the residual-norm ratio jointly and separately in multimodal pretraining with a randomly initialized decoder, which adds value-norm drain as a third axis beyond the text-only dissociations [9, 10].

3 Setup

3.1 Model and arms

All runs use a 222M-parameter nanoVLM [17], in which a pretrained, trainable SigLIP-B/16 vision encoder [18] feeds a decoder with the SmolLM2-135M architecture [19] through a learned projector (Fig. 1A). The decoder has 30 layers of grouped-query attention, 9 query heads per layer sharing 3 KV heads, so there are 270 (layer, query-head) pairs but only 90 (layer, KV-group) value projections. Section 4.3 depends on that distinction. Each sequence is 49 image tokens as a causal prefix followed by 79 left-padded text tokens, 128 in all. There is no BOS token, so **position 0 is the first image token**.

Four training levers each change one thing, and everything else stays byte-identical across arms. *baseline* is plain softmax attention. *g1gate* adds an elementwise σ-gate on the attention output, zero-initialized, after SDPA, the G1 gate of Qiu et al. [20]. *sigmoid* replaces the softmax with an unnormalized sigmoid, after Gu et al. [6]. *textinit* keeps softmax but loads the pretrained SmolLM2-135M decoder, so it imports whatever first-position structure text pretraining built into it (Section 4.1). The decoder starts from random weights in every other arm, and the vision encoder is pretrained in all four. The first two levers have established sink effects in text-only models. *textinit* has no precedent and is an inheritance control. *sigmoid* also tests the standard account of why a sink forms: softmax makes every attention row sum to one, so a head with nothing to retrieve must park its mass somewhere [6]. One caveat on the gate. Unlike Qiu et al. [20], we initialize it at exactly zero, so it opens at $\sigma(0) = 0.5$ and the arm starts as a half-scale output intervention as well as a gated one. We write "Qiu-style G1 in our zero-initialized variant" throughout (Section 5).

3.2 Data and the two training regimes

The four-arm comparison trains on four curated subsets of `the_cauldron` [21], about 146K images, matched at about 100M tokens per arm. *textinit* stops at 60M, where its signatures have plateaued (Section 4.1). *baseline* and *sigmoid* have two seeds, *g1gate* and *textinit* three. The optimizer is AdamW with weight decay 0.1, following [6], gradient clip 1.0 and a cosine schedule with 3% warmup (Appendix A).

At 100M tokens the 146,731-image pool has been cycled about nine times (1.34M samples), a high visual-epoch count. The RF arm (random-fresh) answers that objection. It re-trains the *baseline* recipe on a fresh FineVision stream [22] to 1B tokens over about 4.6M natural images, at 2.39 effective visual epochs. Each example is seen 2.39 times on average, so RF is low-repetition rather than repetition-free. We estimate the overlap between the fresh pool and the repeated subsets at under 3%, from config-level composition rather than image-level deduplication. RF has one seed. Swapping datasets trades the repetition confound for a domain-shift confound (Section 5).

Training stays healthy in every reported run (held-out losses in Appendix A). Two cautions. The lower *textinit* loss reflects its pretrained decoder, not the lever, so only *baseline*, *g1gate* and *sigmoid* are matched on tokens and initialization, differing in the attention lever alone. And the repeated-data arms show a large train–validation gap (`val_seen` near 0.44 against `val_unseen` near 1.18), the overfitting signal that motivates RF. RF has no `val_seen` split, since at 2.39 visual epochs its loader would re-use the held-out pool, so we report `val_unseen` only. It ends at 0.638 with a negative fitted slope over the second half, individual evaluations fluctuating (Section 5).

3.3 Three signatures, tracked separately

We log the three sink symptoms the text-LM literature reports together, each at its own granularity, following Gu et al. [6] at fixed sequence length. Let $a_{l,h}$ be the mean attention that head (l, h) sends to position 0, and $\|v_i^{(l)}\|$ and $\|h_i^{(l)}\|$ the value-vector and residual-stream norms at position i in layer l, all averaged over valid query positions and the fixed probe batch. An overline denotes the mean over the other valid positions $i > 0$. The decoder has $L = 30$ layers, $H = 9$ query heads and $G = 3$ KV groups.

Concentration is the share of (layer, query-head) pairs whose attention to position 0 exceeds a threshold, the metric of [6],

$$\mathrm{Sink}_1^{\epsilon} = \frac{1}{LH} \sum_{l=1}^{L} \sum_{h=1}^{H} \mathbf{1}[a_{l,h} > \epsilon] \, .$$

We use their $\epsilon = 0.3$ default and check $\epsilon \in \{0.2, 0.4\}$. Cross-arm tables report the stricter $\epsilon = 0.2$, which makes an absence claim harder to pass.

Value-norm ratio compares the value norm at position 0 with the rest of the sequence,

$$\text{v-ratio} = \frac{1}{L} \sum_{l=1}^{L} \frac{\|v_0^{(l)}\|}{\overline{\|v_i^{(l)}\|}},$$

below 1 under value-drain [4] and above 1 under amplification. A value vector is shared by the 3 query heads of its KV group, so the ratio rests on $LG = 90$ distinct projections, not 270 (Section 4.3).

Residual-norm ratio is the same ratio on the residual stream,

$$\text{h-ratio} = \frac{1}{L} \sum_{l=1}^{L} \frac{\|h_0^{(l)}\|}{\overline{\|h_i^{(l)}\|}}.$$

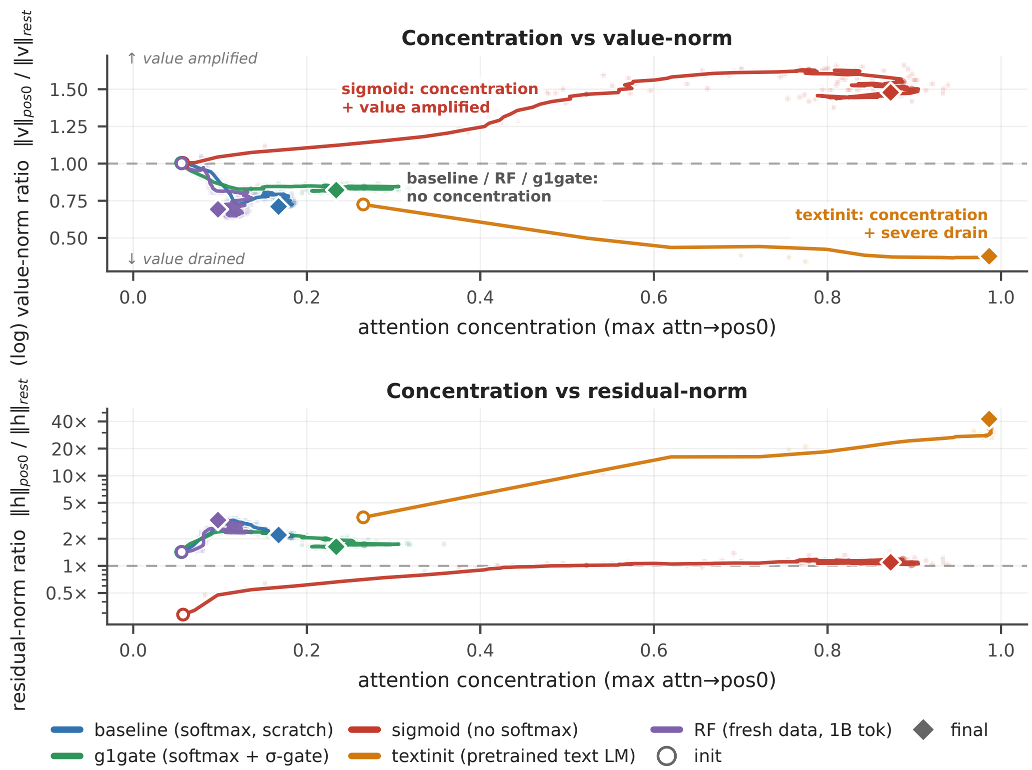

Figure 2: **Decoupling phase-portrait.** Every arm in concentration-against-value-norm space (top) and against the residual-norm ratio (bottom, log scale). Circles mark initialization, diamonds the final checkpoint (about 100M tokens, 60M for *textinit*, 1B for *RF*), seed 0 unless labelled. The horizontal axis is the *maximum* attention→pos0 over heads, not Table 1's fraction, so an arm can sit off zero here with $\mathrm{Sink}^{\epsilon} = 0.000$. Smoothed paths, raw end markers (Appendix F). Were the three signatures one phenomenon, these trajectories would move along one direction.

We call it a massive-activation proxy, since the literature defines massive activations by channel-level outliers [2, 5], which we never measured (Section 5).

All three anchor on position 0 by construction, and we check that choice. At seed 0, per-position attention mass makes position 0 the maximum-mass token in every arm (Appendix E), and Section 5 covers the seed-level exception. The *sigmoid* arm reports the row-normalized attention view, which keeps concentration comparable across arms, and we also log the raw sigmoid mass (Appendix F).

A probe recomputes all three from the live weights every 100 optimizer steps, on the same 32-sample batch for every run and seed, and validates itself against the real forward pass on every call (Appendix A).

4 Results

4.1 Each lever lands in a different corner of signature space

Table 1 collapses the seeds into per-arm ranges, each arm at its final matched checkpoint, about 100M tokens or 60M for *textinit*, whose signatures have plateaued by then (per-seed table in Appendix D). No two arms share a signature triple.

The value-norm axis alone takes three directions, drained hard, drained mildly and amplified, and the corners separate pairwise on single axes. *g1gate* differs from *baseline* on the value-norm axis alone: concentration absent or near-absent and the residual-norm ratio moderate in both, while the gate makes the mild drain milder still, a 15–19% drain rather than none. *sigmoid* and *textinit* both reach strong concentration and then part on the direction of the value-norm move and on the residual-norm

Table 1: **The four corners (n = 2–3 seeds per arm).** Concentration is $\mathrm{Sink}_1^{0.2}$, the fraction of the 270 (layer, query-head) pairs whose mean attention→pos0 exceeds 0.2. The v-ratio rests on 90 (layer, KV-group) value projections and the h-ratio on 30 layers (Section 3).

arm	concentration	value-norm	massive-activation proxy
baseline	absent (0.000)	mild drain (0.69–0.72)	moderate (1.7–2.2)
g1gate	near-absent (0.004–0.011)	**milder drain** (0.81–0.85)	moderate (1.7–2.2)
sigmoid	strong (0.76–0.83)	**amplified** (1.48–1.60)	no strong asymmetry (1.1–1.3)
textinit	strong (0.56–0.85)	strong drain (0.38–0.63)	extreme (5.5–42.5)

Table 2: **RF (fresh stream, n = 1 seed), init → 1B tokens.**

signature	@ init	@ 1B	net
$\mathrm{Sink}_1^{0.3}$ (concentration)	0.000	0.000	flat zero, entire run
max attn→pos0	0.056	0.098	≈flat, far below threshold
h-ratio (massive-activation proxy)	1.43	**3.22**	≈**2.3**×, positive long-horizon trend
v-ratio (value-norm)	1.00	0.69	net drain, non-monotone

ratio. The *sigmoid* corner is relative, since its concentration is row-normalized over a shrinking raw gate budget, with top-head raw pos0 mass 0.065 at the final checkpoint (Appendix F). The arms are not a factorial design, so these are intervention-associated profiles rather than isolated causal effects. The trajectories in Figure 2 separate early and do not share one origin. *textinit* starts from a pretrained decoder that already carries a subthreshold first-position bias, $\mathrm{Sink}_1^{0.3}$ still 0.000 at step 0.

Reproducibility differs by arm. *g1gate* is tightest, $\mathrm{Sink}_1^{0.2}$ of 0.004, 0.011 and 0.0037 across three seeds. The *textinit* corner repeats across seeds and its magnitudes do not: seed 0 sits far above the other two on every signature, h-ratio 42.5 against 5.5–12.2, partly because at seeds 1 and 2 its peaks move off position 0, where our metrics anchor (Appendix D, I). We therefore report its massive-activation proxy as a range, 5.5–42.5× with a median near 12×, and treat the corner rather than any magnitude as the claim.

The corners also separate in time (Fig. A4, seed 0, Appendix I). *baseline*, *g1gate* and *RF* cross the norm thresholds without ever crossing concentration. *sigmoid* is the mirror image, crossing concentration without either norm threshold. *textinit* starts above both norm thresholds and crosses concentration later. Where a run crosses both kinds, the norm signatures come first, and under text initialization the three need not even settle on the same token (Appendix I). Appendix G offers untested hypotheses for why each lever moves a different axis. Figure 3 shows the separation inside a single head. Its rightmost panel carries the most weight: the textinit stripe is already there at step 0, imported with the text-LM weights before the model has seen one image.

4.2 Over 1B fresh tokens the massive-activation proxy more than doubles while concentration stays at zero

The four-arm comparison reuses a 146K-image pool, so its "concentration never emerges" result could be an overfitting artifact. The RF arm answers that with the identical *baseline* recipe on a fresh FineVision stream to one billion tokens at 2.39 effective visual epochs, low repetition rather than none, with a held-out loss that ends at 0.638 on a negative fitted slope over the second half while individual evaluations fluctuate (Sections 3 and 5). Concentration never arrives. $\mathrm{Sink}_1^{0.3}$ **= 0.000 across the entire 0 → 1B run**, and no head of the 270 crosses the threshold at any of about 700 probes.

Warmup does not explain the rise. The h-ratio climbs from 2.40 at about 57M tokens to 3.22 at 1B, long after the 3% warmup window closes, though not monotonically at probe resolution, hence a positive long-horizon trend. In Figure 2, right, the violet trajectory climbs and never moves rightward. The v-ratio ends below its initialization value, but about 75% of that drop happens in the first 57M tokens and part then recovers, so it is supporting context rather than ongoing emergence.

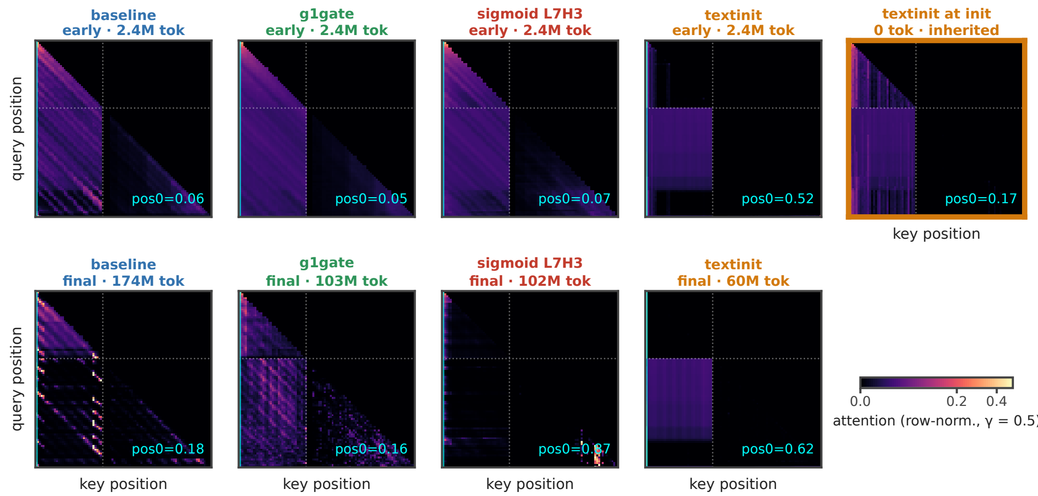

Figure 3: **The sink stripe.** Query × key attention of each arm's top sink head, row-normalized, seed 0. Top row is early training (step 250, about 2.4M tokens), bottom row the final checkpoint. The cyan line marks key = pos0, the dotted line the image-to-text boundary. The stripe is absent in baseline throughout, strong in textinit (attn→pos0 = 0.62 at its final checkpoint), and already present at step 0 in textinit (rightmost panel), inherited from the text LM. The sigmoid heat maps come from an auxiliary streaming batch, while every printed sigmoid number comes from the fixed probe batch (Appendix F).

Table 3: **R(attn→pos0, v-ratio), final checkpoint, seed 0.**

baseline	g1gate	sigmoid	textinit	RF	pooled
+0.76	+0.57	−0.03	−0.79	+0.51	−0.20

4.3 No consistent-sign head-level concentration–value-norm relationship

A tight coupling between concentration and value-drain should at least hold the sign of their per-head correlation constant across regimes. It does not. Table 3 gives the Pearson r between attention→pos0 and value-norm ratio at each arm's final checkpoint, seed 0, over the 90 (layer, KV-group) observations, averaging a group's query-head attention rather than triplicating its value observation (Section 3, scatter in Fig. A2, Appendix C).

These correlations are descriptive, not inferential. Heads within a layer are not independent and each arm rests on a single seed here, so a p-value would mislead and we report none. Collapsing to the 90 KV groups removes the pseudoreplication a per-query-head reading would introduce, and the picture does not change (Appendix F). The pattern is about sign, which pseudoreplication does not manufacture. Heads attending more to position 0 have larger value norms in the baseline regime and smaller ones under text initialization, and the pooled correlation is weak only because arms of opposite sign cancel. We claim that no consistent-sign relationship holds across arms, not that no coupling law exists. A fixed coupling would show one sign everywhere, and it does not.

5 Limitations

Across the arms the three signatures land in different corners, and no single one predicts the others. For VLM work the consequence is direct. An intervention judged on concentration alone can leave value-norm drain or the residual-norm ratio where they were, and a model with no concentration sink can still grow a residual-norm asymmetry, as RF does. What that licenses depends on the five limits below. Appendix B expands each with its evidence, and Section 6 turns the ones we can close into next steps.

The h-ratio is only a proxy The literature defines massive activations by channel-level outliers [2, 5]. We measured a position-specific residual-norm ratio and never computed channel-level statistics (Section 3). A large h-ratio is consistent with massive activations without establishing them.

RF is one seed and one run The 1B-token result rests on a single seed with one weights-only optimizer restart at about 57M tokens, the weights reloaded and the AdamW moments discarded. The audit verified continuity across that seam (Appendix B). Concentration was reproducibly zero across both repeated-data baseline seeds, which supports the negative claim, and a second fresh-data seed would strengthen it. RF also has no seen split, and it buys low repetition by changing dataset, so it trades the repetition confound for a domain-shift one. The under-3% overlap figure is config-level, not image-level.

The gate arm carries a scale confound Zero-initializing the G1 gate opens it at $\sigma(0) = 0.5$ and halves attention output at step 0, where Qiu et al. [20] use ordinary initialization. Gating and initial output scaling are confounded in *g1gate*, and a scale-matched control is future work.

No arm is fully from scratch, and the metrics anchor on position 0 The SigLIP encoder is pretrained in every arm, and vision transformers grow high-norm tokens of their own [25], so part of the residual-norm signal could be inherited. Our defense is the trajectory. The h-ratio starts at 1.0–1.4 and rises to 3.22 across 1B tokens in RF, where a static inherited source predicts a high, flat value from step 0. Position 0 is the maximum-mass token in every arm at seed 0 and at every seed for the random-decoder arms, but not for *textinit* at seeds 1 and 2, where the peaks move (Appendix E). Those magnitudes understate the arm's peaks, so we report *textinit* as a corner and a range, 5.5–42.5×, and not as a magnitude.

Scale and scope Runs reach at most 1B tokens against roughly 5B canonical in the text-LM sink literature [6]. Text-LM sinks form near step 1000 [7], well inside our range, but a signature absent at 1B could still emerge later. The seed-0 probes for the four-arm comparison come from a checksummed archive summary, and seeds 1 and 2 were re-derived first-hand (Appendix B.5, Appendix H). We ran no downstream benchmark on any arm and make no capability claim.

6 Conclusion

In text language models the three attention-sink signatures arrive together. In a vision–language model with a randomly initialized decoder, they need not. Tracked separately, they come apart differently under each training lever applied. Across four interventions the arms occupy separate corners of signature space, with value norms strongly drained, mildly drained or amplified. The sharpest case is the fresh-data run, where over one billion tokens the massive-activation proxy more than doubles while attention concentration stays at exactly zero. This extends the two-way dissociations of text-only models [9, 10] to a separately measured third axis, value-norm drain, in multimodal pretraining. Massive activations can causally produce sinks in text LMs [7]. Here the coupling can also fail to form.

Practical implications The takeaway for VLM deployment and evaluation is that no single signature acted as a reliable proxy for the others in our arms. A model without a concentration sink can still develop a growing residual-norm asymmetry, and an intervention judged successful by one metric may leave the other signatures unchanged. A sink mitigation should therefore report all three.

Status and next steps This is work in progress at small scale, and Section 5 names the gaps. Next steps are a randomly initialized vision encoder to isolate the decoder's contribution to the residual-norm signal, a second fresh-data seed and fresh-data runs past 1B tokens, a scale-matched gate control to separate gating from the half-scale initialization confound, and channel-level statistics to turn the h-ratio proxy into a measurement of massive activations.

Reproducibility A self-validating probe (Section 3) computes all signatures on a fixed probe batch from logs taken every 100 steps, and the Appendix holds the per-seed tables. We will release the code, the probe, the run configurations, the per-run logs and the training checkpoints upon acceptance. The links are withheld here for double-blind review.

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A Extended methodology

Probe. The function `probe_sinks()` re-walks the decoder from the live module weights in eager mode, fp32, no gradients, independent of the training path (fused SDPA kernels, autocast, `torch.compile`). Every call validates its hidden states against the real forward pass to a relative error below 10^{-2}, so the probe cannot drift from what the trained model computes. The probe batch is fixed across all runs and seeds, which keeps every number comparable, and probes fire every 100 optimizer steps.

Recipe. AdamW with weight decay 0.1, following [6], gradient clip 1.0, cosine schedule with 3% warmup. Arms in a comparison differ only in the lever under test.

Optimization. The learning rates are 4×10^{-4} for the language model, 2×10^{-3} for the projector, and 10^{-4} for the vision encoder. We use bf16 autocast, `torch.compile`, and a batch size of 128.

Token accounting. One token is one image token or one non-padding text token. A full sequence contributes 49 image tokens plus its non-padding text tokens, so a step of batch 128 contributes at most 16,384 tokens and about 9.5K on average. All "M tokens" figures in this paper use that definition.

Probe batch. The fixed probe batch holds n = 32 samples, drawn with `random.seed(0)` from the repeated `the_cauldron` tail. We label this probe version `v1-repeatedtail-32`. RF uses the same probe batch as the repeated arms, so its signatures stay comparable to theirs even though its training stream differs (Section 5).

Validation. A 1,024-example held-out pool, evaluated every 500 steps. Each estimate is the mean over the first 512 examples of that pool, in four batches of 128 in fixed order. We report `val_unseen`, which holds out images. For the repeated arms we also log `val_seen`, which re-uses training images with fresh question–answer text. The RF run has no distinct seen split. At 2.39 visual epochs its `val_seen` loader re-uses the held-out pool, so we report only `val_unseen` for RF. At the matched 100M-token checkpoint the held-out losses are 1.182 for *baseline*, 1.133 for *g1gate*, 1.206 for *sigmoid* and 0.877 for *textinit*, and RF reaches 0.638 at 1B tokens. The seed-0 archive values for that checkpoint are 1.161 for *baseline*, 1.138 for *g1gate*, 1.108 for *sigmoid*, and 0.832 for *textinit*, so the first set is seed 1 or 2. For RF the held-out loss goes from 1.35 over the first ten evaluations to 0.68 over the last ten, and the fitted slope over the second half of the run stays negative.

LLM assistance. Large-language-model coding assistants were used to write and refactor the training, probe and analysis code, and to edit prose. The experimental design, the measurements, the analysis decisions and the claims are the authors' own. The released repository keeps its commit history, including the assistant-authored commits.

Aggregation. Each per-layer or per-head quantity is first averaged over the probe batch and over valid query positions, then aggregated by the formulas of Section 3.

B Extended limitations

Section 5 states each limit in one paragraph. This appendix keeps the evidence behind each.

B.1 Metrics anchored on position 0

We measure all three signatures at the first image token, and check that anchor by per-position attention *mass* rather than argmax alone. Position 0 is the maximum-mass token in every arm at seed 0, and it stays so for *baseline*, *g1gate* and *sigmoid* at every seed we scanned. It does not in *textinit*, where at seeds 1 and 2 the peaks move to other positions (per-position table in Appendix E). The pos0-anchored textinit magnitudes at those seeds therefore understate its peaks, one more reason to treat that arm's corner rather than its magnitudes as the claim. In *RF* the attention argmax sits at pos1, but with mass 0.053 against 0.044 at pos0, a diffuse profile rather than a displaced sink, so the RF result does not depend on the anchor.

B.2 The gate arm carries a scale confound

We initialize the G1 gate at exactly zero, so it opens at $\sigma(0) = 0.5$ and halves attention output at step 0, where Qiu et al. [20] use ordinary initialization (Section 3). Gating and initial output scaling are therefore confounded in *g1gate*. We cannot attribute its differences from baseline to gating alone, and a scale-matched control is future work.

B.3 Pretrained vision encoder

The SigLIP encoder is pretrained and trainable in every arm, and *textinit* also uses a pretrained decoder. We study vision–language pretraining with randomly initialized decoders, not from-scratch training of the whole model. Vision transformers grow high-norm register tokens of their own [25], and sinks can propagate from a vision transformer into a vision–language model [11], so part of our residual-norm signal could be inherited rather than decoder-formed. Our defense is the trajectory. The h-ratio starts at 1.0–1.4 and rises to 3.22 across 1B tokens in RF, where inheritance from a static encoder predicts a high, flat h-ratio from step 0. A randomly initialized vision transformer, which would isolate the decoder, is future work.

B.4 Reproducibility of textinit magnitudes

The massive-activation proxy of *textinit* is seed-sensitive, an h-ratio of 5.5–42.5 across three seeds, with seed 0 the outlier on every signature. The reproducible claim is the corner: strong concentration, strong drain, large residual-norm ratio. No specific magnitude is.

B.5 Provenance and seed count

The seed-0 raw probes for the four-arm comparison come from a checksummed archive summary rather than first-hand re-derivation. Seeds 1 and 2 we did re-derive independently, an audit that caught a metric-labeling error in an earlier internal consolidation (Appendix H). *baseline* and *sigmoid* have two seeds, *g1gate* and *textinit* three, RF a single seed. RF also contains one weights-only optimizer restart at about 57M tokens, forced by an out-of-memory error. We reloaded the weights and discarded the AdamW moment estimates, so RF is not one uninterrupted optimizer trajectory. The audit verified continuity across that seam: v-ratio and h-ratio identical at the shared checkpoint, a double-covered 600-step overlap diverging only within probe noise, concentration 0.000 on both sides. Concentration was reproducibly zero across both repeated-data baseline seeds, adequate support for the negative claim, and a second fresh-data seed would strengthen it.

B.6 What the RF control does and does not establish

RF has no distinct seen split, so we cannot run for it the `val_seen` / `val_unseen` comparison that exposes memorization in the repeated arms. The weaker statement its data supports is the falling-slope statement of Section 3. Its streaming shuffle buffer dropped from 1500 to 500 examples partway through, again for memory, so stream ordering is not homogeneous, though the $\mathrm{Sink}_1^{0.3} = 0.000$ result and the h-ratio rise hold within each regime separately. Its probe batch is still the fixed repeated-`the_cauldron` tail used by the other arms. That keeps RF's signatures comparable to theirs, which is what the negative result needs, at the cost of measuring them on data from the other distribution. And RF buys lower repetition by changing dataset, so it trades the repetition confound for a domain-shift one, and we chose the fresh pool to minimize shift. The under-3% overlap figure is config-level, not image-level (Section 3), and the third, domain-matched control we did not run is the known follow-up.

C Supporting figures

D Full per-seed signature table

This table gives the per-seed values behind the collapsed ranges of Table 1, and adds the maximum attention→pos0 where we have it. We logged that metric for seeds 1 and 2 only, because the seed-0 archive summary does not include it. We trust the seed-0 values from a checksummed archive rather

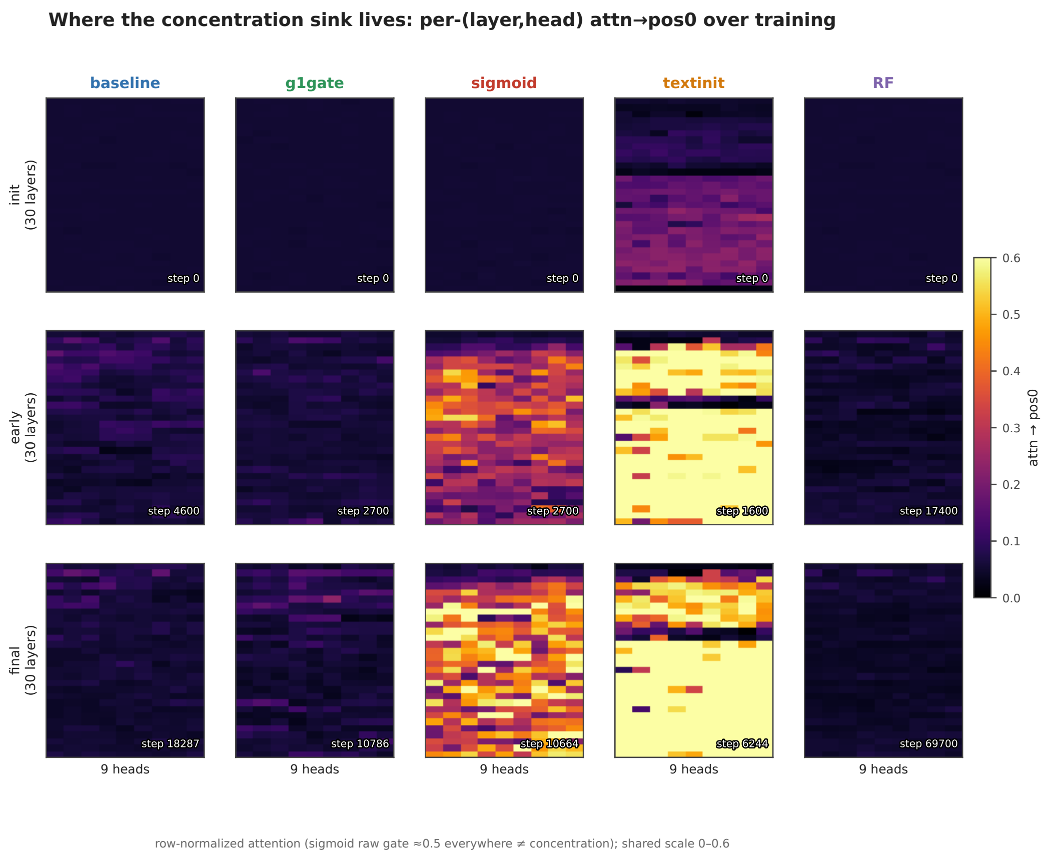

Figure 4: **Where concentration lives in the network.** Mean attention to pos0 for each of the 270 (layer, query-head) pairs through training, row-normalized, seed 0 for every arm, run to each arm's final checkpoint (174M tokens baseline, 103M g1gate, 102M sigmoid, 60M textinit, 1B RF). baseline and RF stay cold throughout. textinit is hot from initialization. sigmoid lights up a band of mid-network layers.

than re-derive them first-hand (Section 5). Each arm is read at its final matched checkpoint. All three signatures of *textinit* have plateaued at 60M tokens, and its h-ratio changes by less than 10% from 60M to 100M in the seeds that continued.

arm	seed	tokens	$\mathrm{Sink}_1^{0.2}$	mean attn→pos0	max attn→pos0	v-ratio	h-ratio
baseline	s0	101M	0.000	0.068	—	0.723	2.16
baseline	s1	100M	0.000	0.062	—	0.687	1.71
g1gate	s0	101M	0.004	0.068	—	0.805	1.67
g1gate	s1	100M	0.011	0.073	~0.21	0.845	2.04
g1gate	s2	100M	0.0037	0.072	~0.22	0.854	2.24
sigmoid	s0	102M	0.830	0.377	—	1.479	1.10
sigmoid	s1	100M	0.756	0.311	—	1.603	1.30
textinit	s0	60M	0.852	0.627	—	0.377	42.5
textinit	s1	60M	0.556	0.235	~0.53	0.634	5.50
textinit	s2	60M	0.578	0.232	~0.59	0.484	12.2

baseline and *sigmoid* are tight across both of their seeds. *textinit* is the loose arm. Seed 0 sits far above the other two on every signature at once, at Sink 0.85 against 0.56–0.58, mean attn→pos0 0.63 against 0.23, and h-ratio 42.5 against 5.5–12.2. The h-ratio has plateaued by 60–100M tokens in both lower seeds, so the spread is genuine seed sensitivity rather than an unconverged transient. Part of it is positional (Appendix E.2).

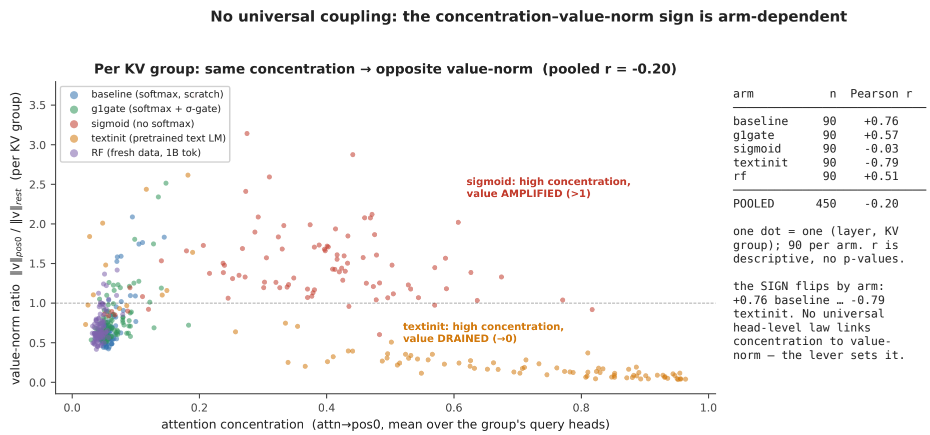

Figure 5: **No consistent-sign head-level relationship.** Concentration against value-norm ratio for each (layer, KV-group) pair at each arm's final checkpoint, seed 0, at n = 90 pairs per arm. Grouping this way avoids triplicating each value-norm observation across the 3 query heads that share it, so these correlations are descriptive and we report no p-values (§3.3). The sign of the correlation flips by arm, from +0.76 in baseline to −0.79 in textinit (Table 3). The pooled cloud is weak, at −0.20, only because arms of opposite sign cancel. Do not read it as an uncorrelated cloud. Most individual arms show a strong |r|.

A note on g1gate. Both seeds with max-attention data show one head at about 0.21–0.22 maximum attention→pos0. That single head does clear the strict $\epsilon = 0.2$ threshold, which is why the arm's $\mathrm{Sink}_1^{0.2}$ is small but not exactly zero, and no head comes close to the $\epsilon = 0.3$ default of [6]. The arm mean meanwhile stays near 0.07 and $\mathrm{Sink}_1^{0.2}$ stays far below 0.05. That pattern repeats across seeds.

E Per-position attention mass (seed 0)

This table gives the mean and max-head attention mass by position at seed 0. It confirms that position 0 is the maximum-mass token in every arm, by mass rather than by argmax count alone. It is the direct check behind the position-0 anchoring defense of Section 5.

arm	mass@pos0 (mean)	max-head@pos0	argmax-mass position	reading
baseline	0.06	0.17	0	pos0 max; diffuse (no sink)
g1gate	0.07	0.23	0	pos0 max; diffuse
sigmoid	0.30	0.66	0	pos0 max; broad raw-sigmoid mass
textinit	0.63	0.99	0	pos0 max; razor spike (pos1 = 0.009)

E.1 Per-position scan at the remaining seeds, and for RF

We re-dumped per-position attention mass at the other seeds and for RF, and per-position residual and value norms for all three *textinit* seeds. The table gives the position at which each quantity peaks (for value norms, the position at which the norm is smallest).

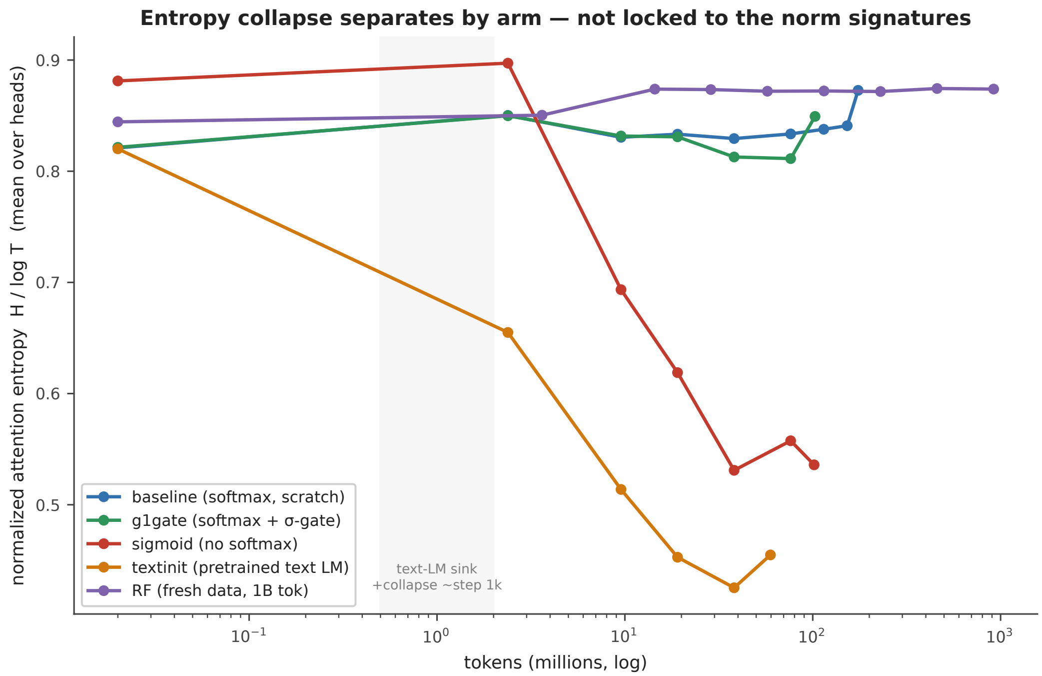

Figure 6: **Entropy collapse tracks concentration only.** Mean attention entropy through training, seed 0, to each arm's final checkpoint. Only sigmoid and textinit collapse. baseline, g1gate, and RF stay flat. The entropy-collapse correlate from the text-LM literature therefore follows the concentration axis, not the norm axes.

run	seed	attention-mass argmax	residual-norm peak	value-norm minimum	reading
baseline	s0, s1	0	—	—	pos0 max
g1gate	s0, s1, s2	0	—	—	pos0 max
sigmoid	s0, s1	0	—	—	pos0 max
textinit	s0	0	0	0	all three coincide
textinit	s1	0	1	5	norms move off pos0
textinit	s2	**1**	**13**	**13**	attention separates from co-incident norm extrema
RF	s0	1	—	—	mass 0.053 vs 0.044 at pos0, diffuse, not a sink

Two readings follow. Position 0 remains the maximum-mass token in every arm with a randomly initialized decoder, so the pos0 anchoring holds where the central negative result lives. In *textinit* the three signatures need not share a token, which is the positional dissociation of Appendix I and the reason the pos0-anchored textinit magnitudes at seeds 1 and 2 understate that arm's peaks.

F Measurement and rendering details

Figure 2 smoothing Paths are smoothed with a moving average, 5 points on the horizontal axis and 9 on the vertical. The faint dots behind each path are the unsmoothed per-probe values, and the initialization and final markers use raw values.

Correlations at the uncollapsed 270 pairs Table 3 collapses to the 90 (layer, KV-group) observations. At the uncollapsed 270 (layer, query-head) pairs the same quantities read +0.67 for baseline, +0.53 for g1gate, -0.03 for sigmoid, -0.76 for textinit, and +0.43 for RF. Every sign and every ordering survives the collapse.

Raw against row-normalized sigmoid attention Row-normalization changes the object being measured, and Gu et al. [6] state their result for *unnormalized* sigmoid attention. Head L7H3 of the sigmoid arm sends 0.873 of its row-normalized attention to position 0 at the final checkpoint, which is the arm maximum. Its raw gate mass to position 0 is 0.065, and the raw mass summed over all keys in that row is 0.110. The row does not sum to one, and pos0 takes about 59% of what little gate mass the head opens at all. Early in training the same head shows the opposite picture: raw pos0 mass 0.309 against a raw row sum of 6.44, so under 5% of a very large gate budget. The concentration this arm develops is therefore a relative reallocation of a shrinking gate budget onto position 0, not the growth of a large absolute mass there (Appendix I). Raw and row-normalized values here come from the same fixed probe batch as every other number in this paper, masked to valid query positions.

Sigmoid heat-map provenance The sigmoid column of Figure 3 uses head L7H3 (0-indexed), the arm's top sink head on the fixed probe batch. An earlier dump selected heads by raw, unnormalized gate score, picked different heads, and understated the arm. The sigmoid heat maps are re-rendered from an auxiliary streaming batch, because the saved matrices covered the superseded heads. Every printed sigmoid number, in the text and in the tables, comes from the fixed probe batch. The pos0 share printed on each panel is computed over valid query positions, the same convention as the tables.

G Interpretation (untested hypotheses)

This appendix is **speculative**. Nothing in it is tested by our experiments, and none of it is a claim of this paper. We include it because a reader is entitled to ask *why* the signatures come apart, and because these hypotheses are cheap to state and testable later.

Gu et al. [6] account for the sink as a key bias. Softmax must distribute a full unit of attention mass per row, and a head with nothing informative to retrieve parks the surplus on a token whose value contributes little. If that account is right, each of our levers touches a different part of it, which would explain why each moves a different axis.

- *g1gate.* An output gate can suppress whatever the attended position injects into the residual stream. A model with that gate may be able to *afford* concentration, because concentration no longer forces a matching change in the value path. That would be consistent with a gate whose measured effect here is on the value-norm axis rather than the concentration axis. Fesser et al. [23] make a compatible argument from another direction. They distinguish two kinds of sink. An "adaptive nop" head suppresses its own update by routing attention to a token whose value contributes nothing, and a negligible value norm identifies it. A broadcast head instead spreads global information. They note that gating implicitly assumes the nop mechanism. If that is right, a gate should act on the value-norm axis first, which is where our gated arm differs from baseline.
- *sigmoid.* Removing the softmax removes the sum-to-one constraint itself. A head with nothing to retrieve can simply attend weakly to everything, with no surplus to park. On this reading the value amplification we observe is what the value path does when it is no longer compensating for a forced allocation.
- *textinit.* A pretrained text decoder arrives with sink machinery already built. Our observation that its norm signatures are present at step 0 while concentration crosses later would then reflect import of structure followed by re-targeting onto a visual prefix, rather than formation from scratch.

Each of these is a hypothesis about mechanism. Testing them needs interventions we did not run: gate-scale sweeps that separate gating from the half-scale confound of Section 3, sigmoid runs at

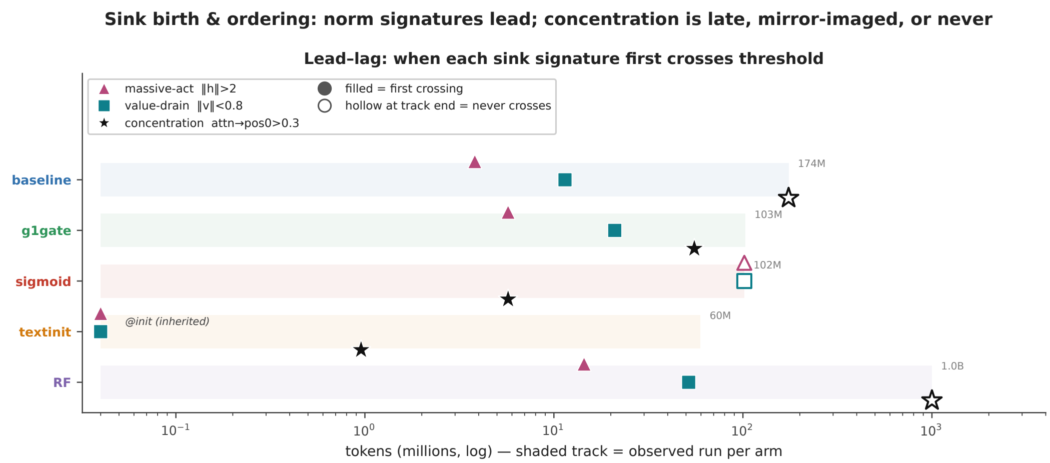

Figure 7: **When each signature first crosses threshold.** Seed 0 throughout. Time-to-event tracks spanning the tokens over which we observed each arm (60M to 1B). A filled marker is the first probe at which a signature crossed its threshold (h>2, v<0.8, attn→pos0>0.3). A hollow marker at a track's end means it never crossed. **Crossing times are interval-censored** at the 100-step probe cadence (head-level map in Fig. A5).

matched effective attention temperature, and text-initialized runs with the sink machinery ablated before alignment.

H Notes on metric hygiene

We record three corrections that we made during the audit of the numbers in this paper. First, an earlier internal consolidation mixed two metrics in one column, mean and max attention→pos0, and thereby manufactured an apparent seed anomaly in *g1gate*. Re-derived like-for-like, the anomaly disappears, and g1gate becomes the most reproducible arm. Second, an earlier draft claimed that the residual-norm peak of textinit "stays pinned at pos0." We checked that claim against the norm dump. It is wrong. The $\|h\|$ peak sits at pos0, pos1 or pos13, depending on the seed (Appendix E.2). That is the positional dissociation reported in Appendix I, which states the corrected form. Third, the per-position script normalized the profile over a 20-position display slice rather than the full 128, which inflated the reported RF masses to 0.100 and 0.083. The true full-sequence values are **0.053 at pos1 and 0.044 at pos0**. The diffuse-profile reading is unchanged, and the softmax-arm rows of Appendix E were unaffected, because those profiles already sum to one over the full sequence.

I I. ordering, the sigmoid measurement note, and positional dissociation

This appendix holds the detail behind the ordering paragraph of Section 4.1.

Timings In the softmax arms with randomly initialized decoders, *baseline* and *g1gate* and *RF*, the residual-norm ratio crosses its threshold within the first few to about 15M tokens and value-drain follows within about 50M. Concentration never comes. No baseline or RF head ever crosses it, and g1gate reaches 1% of heads, a single-layer blip late in training. *sigmoid* mirrors that pattern, with concentration crossing at about 6M tokens and 89% of heads eventually, against 87% for *textinit*, while neither *sigmoid* norm signature ever crosses (Fig. A5). *textinit* inherits its norm signatures rather than growing them. Value-drain and an elevated residual-norm ratio are both present at 0 tokens, imported with the pretrained text-LM weights. Concentration is *not* inherited the same way, since $\mathrm{Sink}_1^{0.3}$ is 0.000 at step 0 and crosses before 1M tokens. Even in the arm that imports the most sink structure, the norm signatures precede concentration. Crossing times are interval-censored at the 100-step probe cadence, so two signatures that cross within one interval should not be read as ordered.

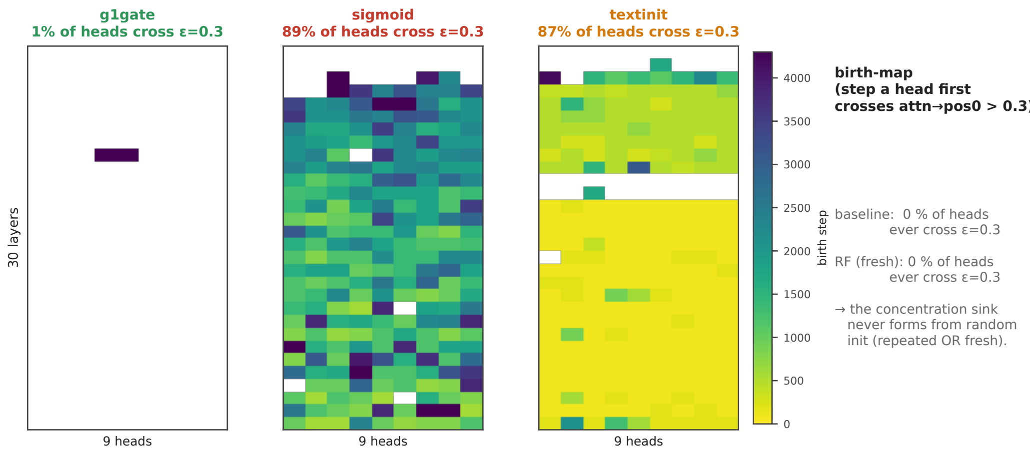

Figure 8: **Birth-maps.** The step at which each (layer, query-head) pair first crosses the concentration threshold (attn→pos0 > 0.3), seed 0, for the three arms in which any head crosses. No *baseline* and no *RF* head ever crosses. 1% of *g1gate* heads do, against 89% for *sigmoid* and 87% for *textinit*. This is the head-level view behind Fig. A4.

The sigmoid arm needs one measurement note, because row-normalization changes the object being measured and Gu et al. [6] state their result for *unnormalized* sigmoid attention. The concentration this arm develops is a **relative** reallocation of a shrinking gate budget onto position 0, not the growth of a large absolute mass there (Appendix F). We report the row-normalized view in the tables so the arms stay comparable.

Entropy collapse Attention-entropy collapse, the text-LM literature's usual correlate of concentration, separates the same way: only *sigmoid* and *textinit* collapse (Fig. A3). It tracks the concentration axis, not the norm axes.

The signatures also dissociate in position A per-position scan across all three *textinit* seeds shows they need not share a token. At seed 0 they coincide at position 0. At seed 1 the attention maximum stays there while the residual peak moves to pos1 and the value minimum to pos5. At seed 2 they separate furthest, attention at **pos1** and both norm extrema at **pos13**. The arms with randomly initialized decoders behave differently: position 0 stays the maximum-mass token in *baseline*, *g1gate* and *sigmoid* at every seed scanned, and in *RF* the attention argmax sits at pos1 with mass 0.053 against 0.044 at position 0, a diffuse profile rather than a sink (Appendix E).

We report this as a supporting observation, not a second headline. It has one consequence for measurement. All three metrics anchor on position 0, so the *textinit* magnitudes at seeds 1 and 2 are read at a token that is no longer the peak and therefore **understate** the arm's true peaks, a further reason to report that arm as a range and a median (Sections 4.1 and 5). The anchoring holds for the randomly initialized decoders, which carry the central negative result.

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   Answer: [Yes]

   Justification: The abstract and Section 1 state exactly the three contributions the paper supports, each with its scope attached: four signature corners at $n = 2$–3 seeds per arm, the low-repetition 1B-token result at a single seed, and the sign-flipping per-head correlation reported at seed 0 and described as descriptive rather than inferential. Section 1 also states plainly that two-way decoupling in text-only models is not ours, and that no downstream behaviour is tested here.

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   Answer: [Yes]

   Justification: Section 5 is a dedicated Limitations section covering position-0 anchoring, the scale confound in the gate arm, the pretrained vision encoder, the h-ratio being a proxy rather than a channel-level measurement of massive activations, token scale, seed-level reproducibility of the textinit magnitudes, provenance and seed counts including the RF optimizer restart, what the RF control cannot establish, domain shift, and the absence of any benchmark accuracy.

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